

GCRew_Flume_TidalSurvey_DOC_QAQC_202308

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2026-04-28

##Setup - Change things here & write any notes

```
#identify section
cat("Setup Information")
```

Setup Information

```
##### Run information - PLEASE CHANGE
Date_Run = "08/25/2023" #Date that instrument was run
Run_by = "Stephanie J. Wilson" #Instrument user
Script_run_by = "Stephanie J. Wilson" #Code user
run_notes = "Standard Curve was assessed manually on the instrument." #any notes from the run
#samples <- c("Flume") #whatever identifies your samples within the same names
samples_pattern <- "Flume" #paste(samples, collapse = "|")
#samples_pattern <- "GCW" #use this instead of the line above if you have only one site code
chks_name = "Chk_Std_50C_2N" #what did you name your check standards?

library(here)
```

Warning: package 'here' was built under R version 4.3.3

here() starts at S:/Biogeochemistry/People/Steph Wilson/Github/serc-lateral-flux

```
##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
#raw_file_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202505.txt" #example
raw_file_name = here("Tidal_Surveys/Raw_Data/202308", "TOCT_Flume_TidalSurvey_Aug2023_TCTN.txt")

#file path and name for raw all peaks file
#raw_allpeaks_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202505_allpeaks.txt" #example
# raw_allpeaks_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202411_allpeaks.txt"

#file path and name of processed data file
#processed_file_name = "Processed Data/COMPASS_SynopticCB_PW_Processed_DOC_202505.csv" #example
processed_file_name = here("Tidal_Surveys/Processed_Data/202308", "Flume_TidalSurvey_DOC_Processed_202308.csv")

##### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
Raw_Metadata = here("Tidal_Surveys/Raw_Data/202308", "Tidal_Survey_202308_Metadata.csv")
```

```
#qac log file path for this year
#Log_path = "Raw Data/COMPASS_Synoptic_TOCTN_QAClog_2024.csv"
```

```
##Set Up Code
```

Import Data Functions

Import Sample Data

```
## Import Sample Data
```

```
## New names:
```

```
## * ' ' -> '...14'
```

```
## # A tibble: 6 x 4
```

```
##   sample_name      npoc_raw tdn_raw run_datetime
##   <chr>          <dbl>    <dbl> <chr>
## 1 Flume_20230807_GW-1      75.5    6.76 8/21/2023 12:35:28 AM
## 2 Flume_20230807_GW-2      38.2    0.892 8/21/2023 1:01:14 AM
## 3 Flume_20230807_GW-3      27.1    0.536 8/21/2023 1:18:25 AM
## 4 Flume_20230807_GW-4      51.1    1.31 8/21/2023 1:44:20 AM
## 5 Flume_20230807_GW-5      36.5    0.922 8/21/2023 2:11:13 AM
## 6 Flume_20230807_GW-6      28.0    0.590 8/21/2023 2:28:10 AM
```

Assessing Standard Curves - assessed manually on the instrument

Assess Check Standards

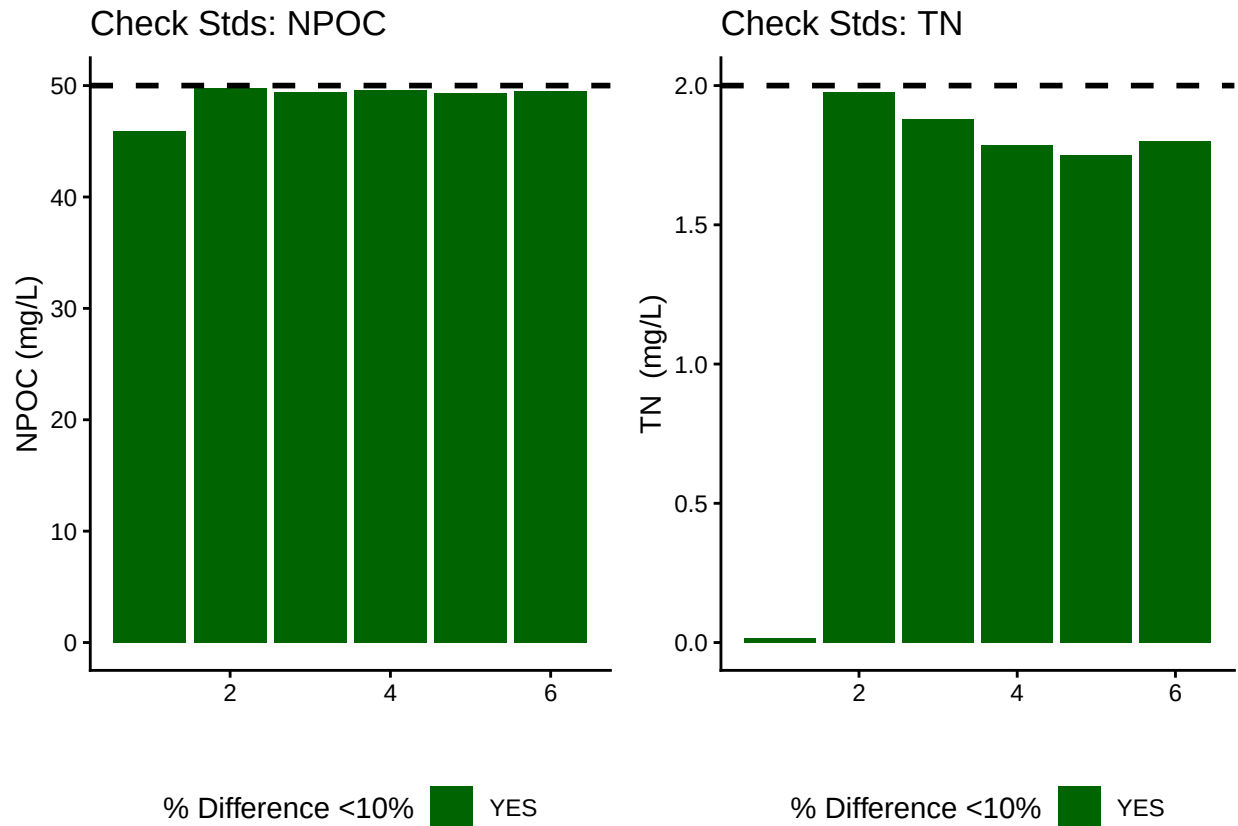
```
## Assess the Check Standards
```

```
## New names:
```

```
## * ' ' -> '...14'
```

```
## [1] "Carbon Check Standard RSD within Range"
```

```
## [1] "Nitrogen CHECK STANDARD RSD TOO HIGH - REASSESS"
```



```
## [1] ">60% of Carbon Check Standards are within range of expected concentration"
```

```
## [1] ">60% of Nitrogen Check Standards are within range of expected concentration"
```

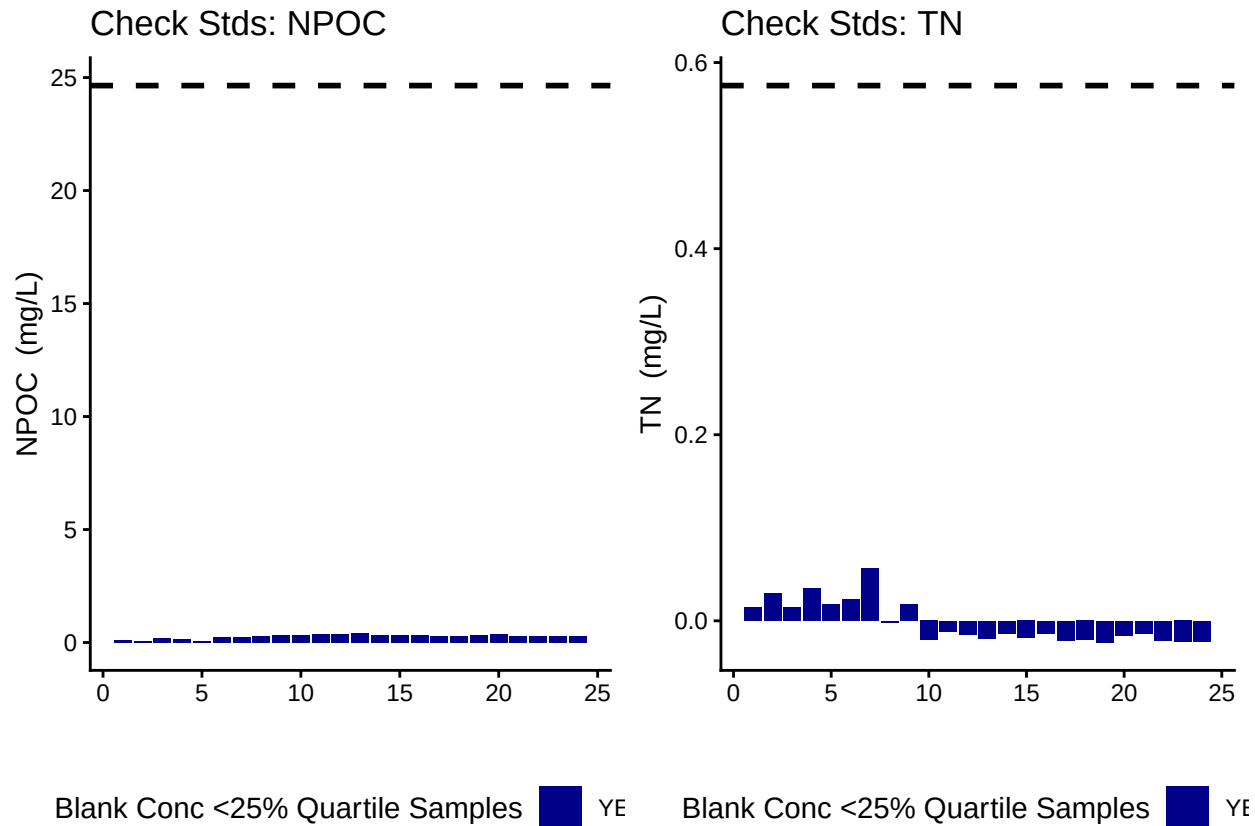
Assess Blanks

```
## Assess Blanks
```

```
## New names:
## * ' ' -> '...14'
```

```
## [1] ">60% of Carbon Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">60% of Nitrogen Blank concentrations are lower 25% quartile of samples"
```



```
## carbon blanks:
```

```
## [1] 0.2819475
```

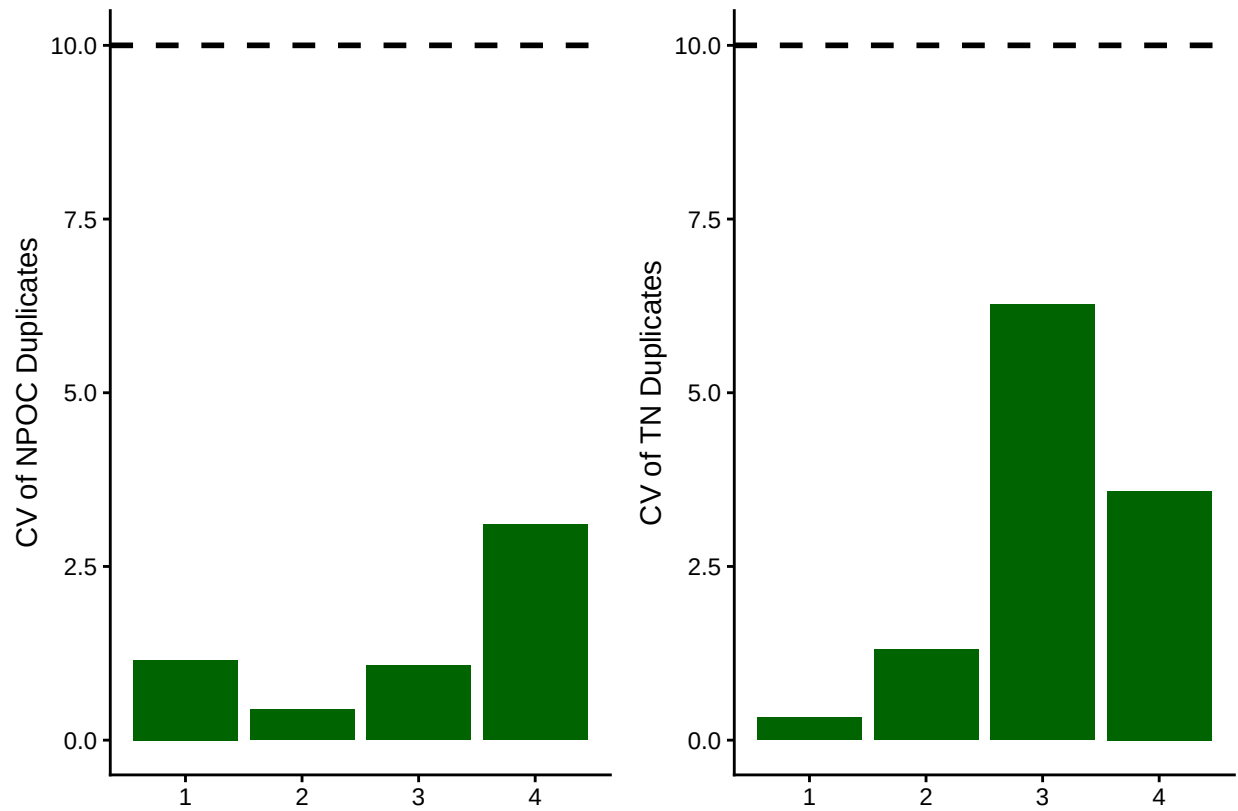
```
## nitrogen blanks:
```

```
## [1] -0.0026875
```

Assess Duplicates

```
## Assess Duplicates
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

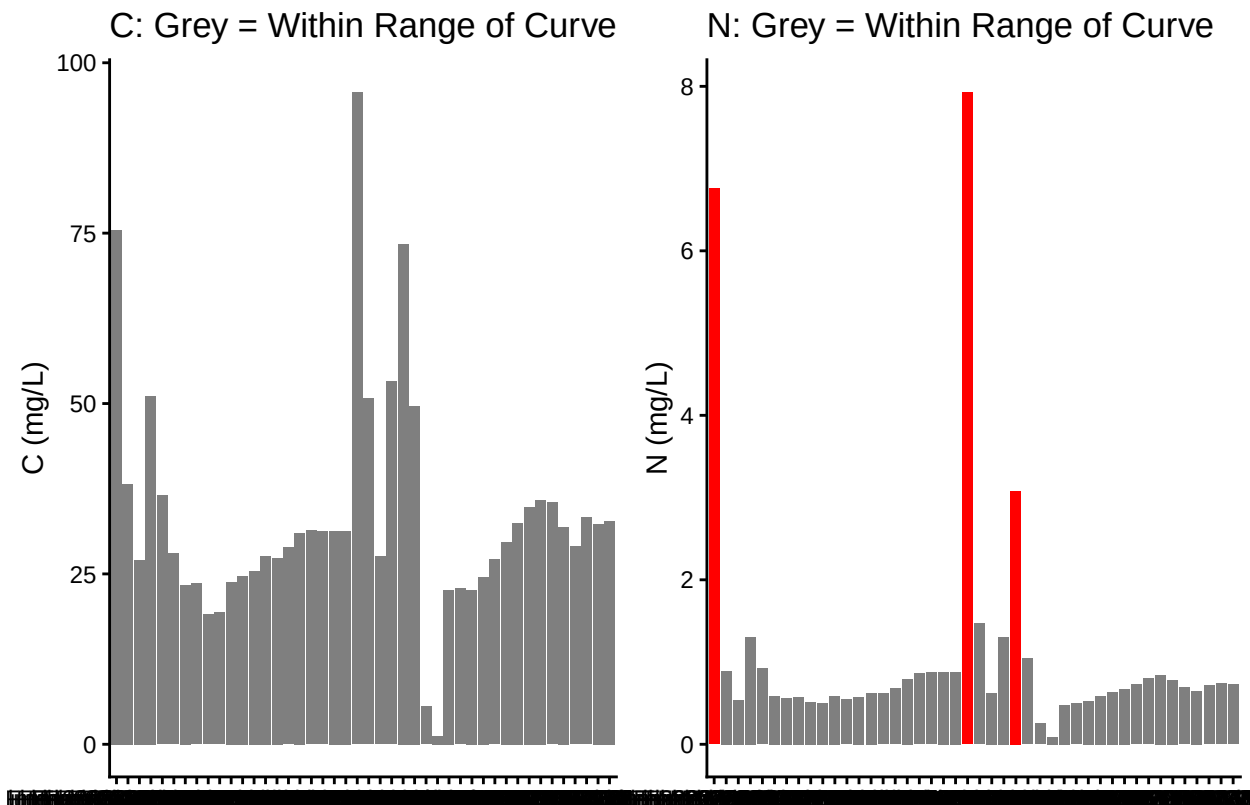


```
## [1] ">60% of Carbon Duplicates have a CV <10%"
```

```
## [1] ">60% of Nitrogen Duplicates have a CV <10%"
```

Sample Flagging - Are samples Within the range of the curve?

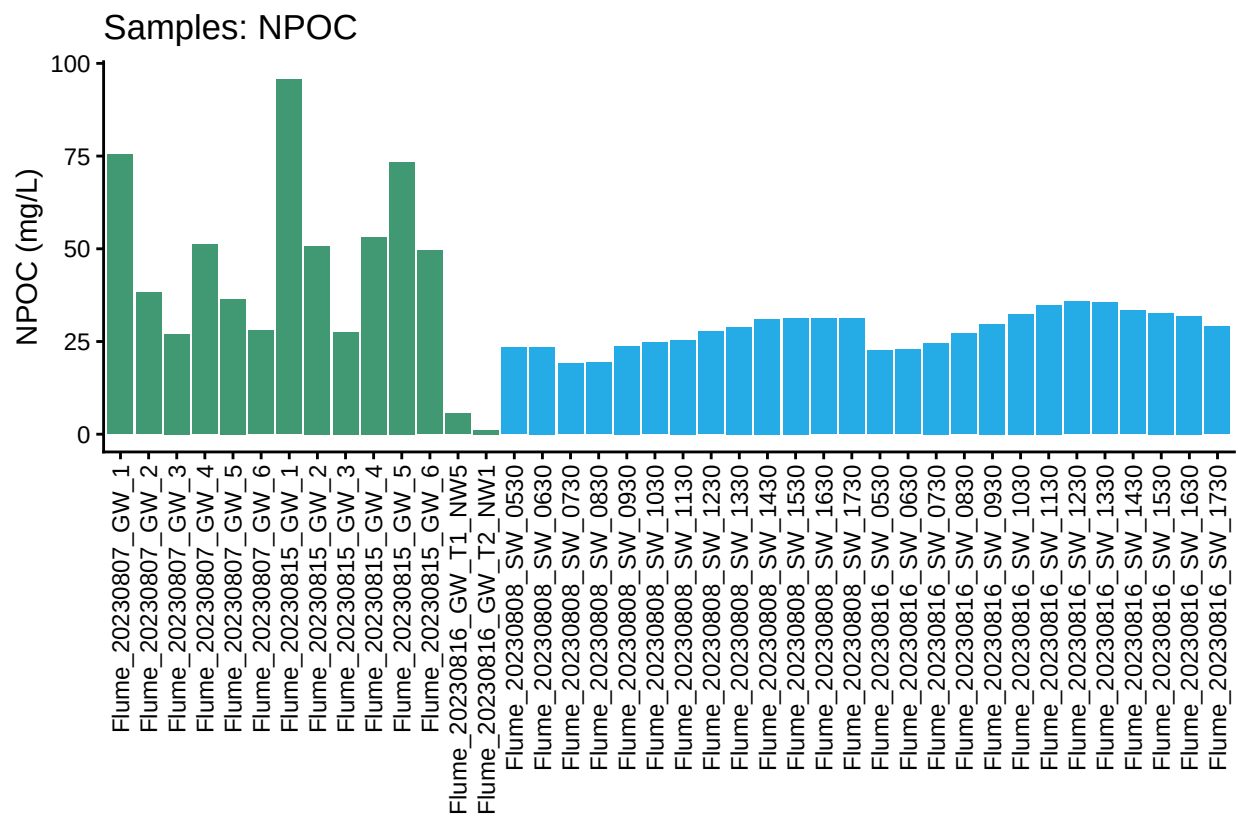
```
## Sample Flagging
```

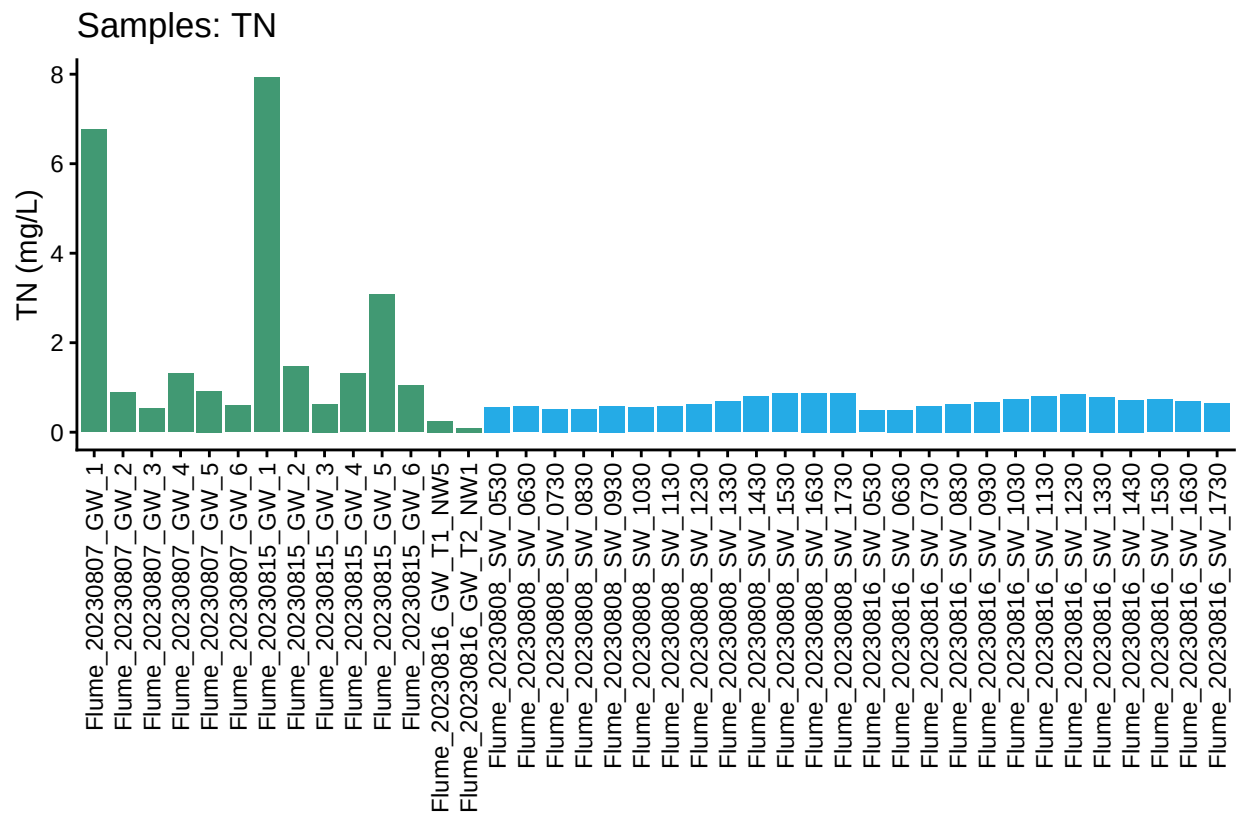


Visualize Data by Plot

```
## Visualize Data
```

```
## Warning in rbind(c("Flume", "20230807", "GW", "1"), c("Flume", "20230807", :
## number of columns of result is not a multiple of vector length (arg 1)
```





Convert data from mg/L to uMoles/L

Match metadata & merge info

Merge with Metadata

##	Survey_Type	Site	Date	Time_EDT	Time_EST	Sample_Type
## 1	Tidal Survey	Flume	8/8/2023	5:30	4:30	SW
## 2	Tidal Survey	Flume	8/8/2023	6:30	5:30	SW
## 3	Tidal Survey	Flume	8/8/2023	7:30	6:30	SW
## 4	Tidal Survey	Flume	8/8/2023	8:30	7:30	SW
## 5	Tidal Survey	Flume	8/8/2023	9:30	8:30	SW
## 6	Tidal Survey	Flume	8/8/2023	10:30	9:30	SW

##	Sample_ID	DO_mgL	pH	Salinity_ppt	Temp	DIC	DICex	SO4_H2S	NUTR	DOC
## 1	Flume_20230808_SW_0530	<NA>	NA	<NA>	NA	x	x	x	x	x
## 2	Flume_20230808_SW_0630	<NA>	NA	<NA>	NA	x	x	x	x	x
## 3	Flume_20230808_SW_0730	<NA>	NA	<NA>	NA	x	x	x	x	x
## 4	Flume_20230808_SW_0830	<NA>	NA	<NA>	NA	x	x	x	x	x
## 5	Flume_20230808_SW_0930	<NA>	NA	<NA>	NA	x	x	x	x	x
## 6	Flume_20230808_SW_1030	<NA>	NA	<NA>	NA	x	x	x	x	x

##	GHGs	TA	Radium_Vol_L	Notes
## 1	x	x	22.5	August: Neap Tide
## 2	x	x	22.5	August: Neap Tide
## 3	x	x	22.5	August: Neap Tide
## 4	x	x	22.5	August: Neap Tide


```
## 5      x      x      22.5 August: Neap Tide
## 6      x      x      22.5 August: Neap Tide
```

```
## All sample IDs are present in metadata.
```

Export Processed Data

```
## Export Processed Data
```

```
## # A tibble: 6 x 22
##   Project Site Replicate Sample_Type Sample_ID Date Time_EDT Time_EST npoc_mgL
##   <chr>   <chr> <chr>      <chr>      <chr>   <chr> <chr>   <chr>      <dbl>
## 1 SERC L~ Flume 1      GW      Flume_20~ 8/7/~ 8:15    7:15    75.5
## 2 SERC L~ Flume 2      GW      Flume_20~ 8/7/~ 8:50    7:50    38.2
## 3 SERC L~ Flume 3      GW      Flume_20~ 8/7/~ 9:20    8:20    27.1
## 4 SERC L~ Flume 4      GW      Flume_20~ 8/7/~ 9:45    8:45    51.1
## 5 SERC L~ Flume 5      GW      Flume_20~ 8/7/~ 10:30   9:30    36.5
## 6 SERC L~ Flume 6      GW      Flume_20~ 8/7/~ 11:10   10:10   28.0
## # i 13 more variables: npoc_uM <dbl>, npoc_flag <chr>, tdn_mgL <dbl>,
## #   tdn_uM <dbl>, tdn_flag <chr>, Analysis_runtime <chr>, Run_notes <chr>,
## #   Field_notes <chr>, DO_mgL <chr>, pH <dbl>, Salinity_ppt <chr>, Temp <dbl>,
## #   Radium_Vol_L <dbl>
```

```
#end
```